

Topology First-Year Exam, August 2023, Part 2

Work the following problems and show all work. Support all statements to the best of your ability.

1. Show that B^2 is not homeomorphic to B^3 , where B^n denotes the closed unit ball in $\mathbb{R}^n$.
2. Show that every continuous map $f : S^2 \rightarrow S^1$ of the 2-sphere to the circle S^1 is null-homotopic.
3. Let

$$X = ([-1, 1] \times \{0\} \times \{0\}) \cup (\{0\} \times [-1, 1] \times \{0\}) \cup (\{0\} \times \{0\} \times [-1, 1]) \subset \mathbb{R}^3.$$

Show that every continuous map $f : X \rightarrow X$ has a fixed point.

4. Derive the Brouwer Fixed Point Theorem for B^2 from the Borsuk–Ulam Theorem for S^2 .
5. Let X , Y , and Z denote the following open rays in $\mathbb{R}^3$:

$$X = (0, \infty) \times \{0\} \times \{0\}, \quad Y = \{0\} \times (0, \infty) \times \{0\}, \quad Z = \{0\} \times \{0\} \times (0, \infty).$$

Compute $\pi_1(A)$ where

- (a) $A = \mathbb{R}^3 \setminus X$?
- (b) $A = \mathbb{R}^3 \setminus \bar{X}$? Here $\bar{X}$ denotes the closure of X in $\mathbb{R}^3$.
- (c) $A = \mathbb{R}^3 \setminus (X \cup Y)$?
- (d) $A = \mathbb{R}^3 \setminus (\bar{X} \cup Y)$?
- (e) $A = \mathbb{R}^3 \setminus (X \cup Y \cup Z)$?
- (f) $A = \mathbb{R}^3 \setminus (\bar{X} \cup Y \cup Z)$?

Answer the following with complete definitions or proofs.

6. Describe a space X with $\pi_1(X) \cong \mathbb{Z}_3$.
7. Show that the projective plane $\mathbb{R}P^2$ is not homeomorphic to the surface of genus 2, $M_2 = T \# T$.
8. Let $X = [0, 1]^{(0,1)}$ be given the product topology. Is the space X
 - (a) compact?
 - (b) metrizable?
 - (c) separable?
9. State the Seifert–van Kampen Theorem. Can it be used to compute $\pi_1(S^n)$?
10. State the Tychonoff Theorem. Prove that every compact Hausdorff space X can be embedded into a Tychonoff cube $[0, 1]^A$.